Pocket SDR: Command Reference

pocket_scan

Synopsis

```
pocket_scan [-e]
```

Description

Scan and list USB devices.

Options

-e Show end point information for USB devices.

Examples

pocket_conf

Synopsis

```
pocket_conf [-s] [-a] [-h] [-p bus[,port]] [conf_file]
```

Description

Configure or show settings for a Pocket SDR FE device. If conf_file specified, the settings in the configuration file are set to the Pocket SDR FE device registers. The configuration is a text file containing records of MAX2771 register field settings as like follows. The register field settings are written as keyword = value format or hexadecimal format. In the case of keyword = value format, a keyword is a field name shown in MAX2771 manual [1]. Strings after # in a line is treated as comments. If conf_file omitted, the command shows the settings of the Pocket SDR FE device in the same format of the configuration file.

```
Keyword = value format:

[CHx]
FCEN     = 97  # ...
FBW      =  0  # ...
F3OR5    =  1  # ...
...

Hexadecimal format:

#CH  ADDR       VALUE
  1  0x00  0xA2241C17
  1  0x01  0x20550288
...
```

Options

- -s

Save the settings to EEPROM of the SDR device. These settings are also loaded at reset of the Pocket SDR FE device.

- -a

Show all of the register fields.

- -h

Configure or show registers in a hexadecimal format.

- -p [bus[,port]]

USB bus and port number of the Pocket SDR FE device. Without the option, the command selects the device firstly found.

- conf_file

Path of the configuration file. Without the option, the command shows current register field settings of the Pocket SDR FE device.

References

[1] maxim integrated, MAX2771 Multiband Universal GNSS Receiver, July 2018

pocket_dump

Synopsis

```
pocket_dump [-t tsec] [-r] [-p bus[,port]] [-c conf_file] [-q]
            [file [file ...]]
```

Description

Capture and dump digital IF (DIF) data of a Pocket SDR FE device to output files. To stop capturing, press Ctr-C.

Options

-t tsec

```
Data capturing time in seconds.
```

-r

```
Dump raw data of the Pocket SDR FE device without channel separation
and quantization.
```

-p bus[,port]

```
USB bus and port number of the Pocket SDR FE device. Without the
option, the command selects the device firstly found.
```

-c conf_file

```
Configure the Pocket SDR FE device with a device configuration file
before capturing.
```

-q

```
Suppress showing data dump status.
```

[file [file ...]]

```
Output digital IF data file paths. The first path is for CH1,
the second one is for CH2 and so on. The second one or the later
can be omitted. With option -r, only the first path is used. If
the file path is "", data are not output to anywhere. If the file
path is "-", data are output to stdout. If all of the file paths
omitted, the following default file paths are used.

CH1: ch1_YYYYMMDD_hhmmss.bin
CH2: ch2_YYYYMMDD_hhmmss.bin
...
(YYYYMMDD: dump start date in UTC, hhmmss: dump start time in UTC)
```

pocket_acq

Synopsis

```
pocket_acq [-sig sig] [-prn prn[,...]] [-tint tint] [-toff toff]
    [-f freq] [-fi freq] [-d freq] [-nz] file
```

Description

Search GNSS signals in digital IF data and plot signal search results. If single PRN number by -prn option, it plots correlation power and correlation shape of the specified GNSS signal. If multiple PRN numbers specified by -prn option, it plots C/N0 for each PRN.

Options ([]: default)

-sig sig

```
GNSS signal type ID (L1CA, L2CM, ...). See below for details. [L1CA]
```

-prn prn[,...]

```
PRN numbers of the GNSS signal separated by ','. A PRN number can be a
PRN number range like 1-32 with start and end PRN numbers. For GLONASS
FDMA signals (G1CA, G2CA), the PRN number is treated as FCN (frequency
channel number). [1]
```

-tint tint

```
Integration time in ms to search GNSS signals. [code cycle]
```

-toff toff

```
Time offset from the start of digital IF data in ms. [0.0]
```

-f freq

```
Sampling frequency of digital IF data in MHz. [12.0]
```

-fi freq

```
IF frequency of digital IF data in MHz. The IF frequency equals 0, the
IF data is treated as IQ-sampling (zero-IF). [0.0]
```

-d freq[,freq]

```
Reference and max Doppler frequency to search the signal in Hz.
[0.0,5000.0]
```

-nz

```
Disalbe zero-padding for circular colleration to search the signal.
[enabled]
```

-h

```
Show usage and signal type IDs
```

file

```
File path of the input digital IF data. The format should be a series of
int8_t (signed byte) for real-sampling (I-sampling) or interleaved int8_t
for complex-sampling (IQ-sampling). PocketSDR and AP pocket_dump can be
used to capture such digital IF data.
If the tag file <file>.tag of the input IF data exists, the format,
the sampling frequency, the LO frequencies and the sampling types
are automatically recognized by the tag file and the options -fmt,
-f, -fi, and -IQ are ignored.
```

pocket_trk

Synopsis

```
pocket_trk [-sig sig -prn prn[,...] [-rfch ch[,...]] ...]
    [-fmt {INT8|INT8X2|RAW8|RAW16|RAW32}] [-f freq] [-fo freq[,...]]
    [-IQ {1|2}[,...]] [-toff toff] [-ti tint] [-p bus,[,port]
    [-c conf_file] [-log path] [-nmea path] [-rtcm path] [-raw path]
    [-w file] [file]
```

Description

It searchs and tracks GNSS signals in the input digital IF data, extract observation data, decode navigation data and generate PVT solutions. The observation and navigation data can be output as a RTCM3 stream. The PVT solutions can be output as a NMEA stream. The observation data and raw navigation data and some event logs can be output as a log stream.

Options ([]: default)

-sig sig -prn prn[,...] [-rfch ch[,...]] ...

```
A GNSS signal type ID (L1CA, L2CM, ...) and a PRN number list of the
signal. For signal type IDs, refer pocket_acq.py manual. The PRN
number list shall be PRN numbers or PRN number ranges like 1-32 with
the start and the end numbers. They are separated by ",". For
GLONASS FDMA signals (G1CA, G2CA), the PRN number is treated as the
FCN (frequency channel number). To assign the signal to specific RF
channel(s), -rfch option can be followed. Specify the assigned RF
channel list with "," or "-". Without the -rfch option, RF channel
for the signal is automatically assigned. The signal option can be
repeated for multiple GNSS signals to be tracked.
```

-fmt {INT8|INT8X2|RAW8|RAW16|RAW32}

```
Specify IF data format as follows: INT8 = int8 (I-sampling), INT8X2 =
interleaved int8 (IQ-sampling), RAW8 = Pocket SDR FE 2CH raw (packed
8 bits), RAW16 = Pocket SDR FE 4CH raw (packed 16 bits), RAW32 =
Pocket SDR FE 8CH raw (packed 32 bits) [INT8X2]
```

-f freq

```
Specify the sampling frequency of the IF data in MHz. [12.0]
```

-fo freq[,...]

```
Specify LO frequency for each RF channel in MHz. In case of the
IF data format as RAW8, RAW16 or RAW32, multiple (2, 4 or 8)
frequencies have to be specified separated by ",".
```

-IQ {1|2}[,...]

```
Specify the sampling type (1 = I-sampline, 2 = IQ-sampling) for each
RF channel separated by "," in case of the IF data foramt as RAW8,
RAW16 or RAW32. [2,2,2,2,2,2,2,2]
```

-toff toff

```
Time offset from the start of the IF data in s. [0.0]
```

-tscale scale

```
Time scale to replay the IF data file. [1.0]
```

-ti tint

```
Update interval of the signal tracking status in seconds. If 0
specified, the signal tracking status is suppressed. [0.1]
```

-p bus[,port]

```
USB bus and port number of the Pocket SDR FE device in case of IF data
input from the device.
```

-c conf_file

```
Configure the Pocket SDR FE device with a device configuration file
before signal acquisition and tracking.
```

-log path

```
A stream path to write the signal tracking log. The log includes
observation data, navigation data, PVT solutions and some event logs.
The stream path should be one of the followings.

(1) local file file path without ':'. The file path can be contain
    time keywords (%Y, %m, %d, %h, %M) as same as the RTKLIB stream.
(2) TCP server  :port
(3) TCP client  address:port
```

-nmea path

```
A stream path to write PVT solutions as NMEA GNRMC, GNGGA and GNGSV
sentences. The stream path is as same as the -log option.
```

-rtcm path

```
A stream path to write raw observation and navigation data as RTCM3.3
messages. The stream path is as same as the -log option.
```

-raw path

```
A stream path to write raw IF data. The stream path is as same as the
-log option.
```

-h height

```
Specify the console height (rows). [64]
```

-w file

```
Specify the FFTW wisdowm file. [../python/fftw_wisdom.txt]
```

[file]

```
A file path of the input IF data. The Pocket SDR FE deveice and
pocket_dump can be used to capture such digitized IF data.

If the tag file <file>.tag for the input IF data exists, the format,
the sampling frequency, the LO frequencies and the sampling types
are automatically recognized by the tag file. In this case, the
options -fmt, -f, -fo, and -IQ are ignored.

If the file path omitted, the input is taken from a Pocket SDR FE
device directly. In this case, the format, the sampling frequency,
the LO frequencies and the sampling types are automatically
configured according to the device information.
```

pocket_snap

Synopsis

```
pocket_snap [-ts time] [-pos lat,lon,hgt] [-ti sec] [-toff toff]
     [-f freq] [-fi freq] [-tint tint] [-sys sys[,...]] [-v] [-w file]
     -nav file [-out file] file
```

Description

Snapshot positioning with GNSS signals in digitized IF file.

Options ([]: default)

-ts time

```
Captured start time in UTC as YYYY/MM/DD HH:mm:ss format.
[parsed by file name]
```

-pos lat,lon,hgt

```
Coarse receiver position as latitude, longitude in degree and
height in m. [no coarse position]
```

-ti sec

```
Time interval of positioning in seconds. (0.0: single) [0.0]
```

-toff toff

```
Time offset from the start of digital IF data in seconds. [0.0]
```

-f freq

```
Sampling frequency of digital IF data in MHz. [12.0]
```

-fi freq

```
IF frequency of digital IF data in MHz. The IF frequency equals 0, the
IF data is treated as IQ-sampling (zero-IF). [0.0]
```

-tint tint

```
Integration time for signal search in msec. [20.0]
```

-sys sys[,...]

```
Select navigation system(s) (G=GPS,E=Galileo,J=QZSS,C=BDS). [G]
```

-v

```
Enable verpose status display.
```

-w file

```
Specify FFTW wisdowm file. [../python/fftw_wisdom.txt]
```

-nav file

```
RINEX navigation data file.
```

-out file

```
Output solution file as RTKLIB solution format.
```

file

```
Digitized IF data file.
```